

Granoblastic Polygonal Texture, Annealing & Grain Boundary Migration

Authoritative Reference: John D. Winter, *Principles of Igneous and Metamorphic Petrology* (2nd Ed., Chapter 23: Structures and Textures of Metamorphic Rocks)

Comprehensive Chapter Masterclass: Metamorphic Textures & Recrystallization

Recovery, Dynamic vs. Static Recrystallization, Interfacial Surface Free Energy & 120° Triple Junctions

1. The Thermodynamic Driving Force for Recrystallization

In metamorphic geology, **recrystallization** is defined as the structural rearrangement of mineral grains without necessarily changing their bulk chemical composition. When rocks undergo deformation, strain energy accumulates within the crystal lattice in the form of **dislocations, subgrains, and high-energy curved grain boundaries**.

When deformation ceases but elevated temperature persists—a process known as **static annealing**—the rock seeks thermodynamic equilibrium by minimizing two primary sources of excess internal free energy:

1. **Lattice Strain Energy:** Minimized during initial *recovery* through subgrain rotation and dislocation climb/glide.
2. **Interfacial Surface Free Energy (γ):** Grains possess excess energy along their mutual contact boundaries. Thermodynamics dictates that the system must reduce total boundary surface area per unit volume: $\Delta G_{\text{surface}} = \sum \gamma_i A_i$. To minimize this surface free energy, high-angle grain boundaries migrate toward centers of curvature, eliminating small strained grains, straightening boundary segments, and achieving an optimal geometric packing.

"In cases of advanced recrystallization, grain boundary area reduction straightens the boundaries further, eventually producing a granoblastic polygonal texture that is completely recrystallized (annealed). The grain boundaries meet at approximately 120° angles, representing the lowest interfacial energy configuration (foam texture)."

— John D. Winter, *Principles of Igneous and Metamorphic Petrology* (2nd Ed., Chapter 23, p. 510)

b recovery produces two strain-free subgrains

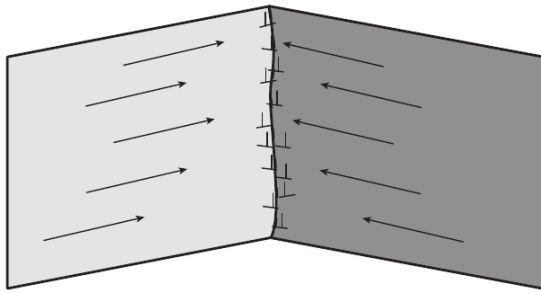


FIGURE 5 Illustration of a recovery process in which dislocations migrate to form a subgrain boundary.

Recrystallization occurs by grain boundary migration or subgrain rotation (Figure 6). In **grain boundary migration** (Figures 6a and 7a), the atoms in a grain with higher strain energy migrate by diffusion and add to a neighboring grain with lower strain energy (also called **bulging**). Less commonly, an isolated new grain may nucleate in a highly

b sub-grain rotation

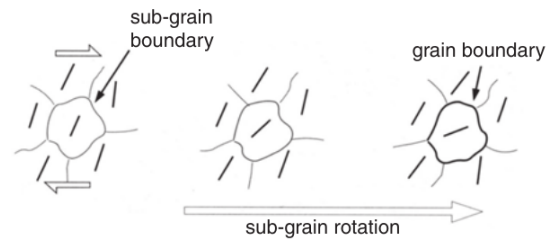


FIGURE 6 Recrystallization by (a) grain-boundary migration (including nucleation) and (b) subgrain rotation. From Passchier and Trouw (2005). Copyright © with permission from Springer-Verlag.

strained area of a deformed grain. **Subgrain rotation** (Figure 6b) occurs when dislocations are free to creep from the lattice plane of one subgrain to that of another and add (almost continuously) to a neighboring subgrain. The added dislocations cause the receiving subgrain lattice to rotate until it is sufficiently different from its neighbors to qualify as a separate

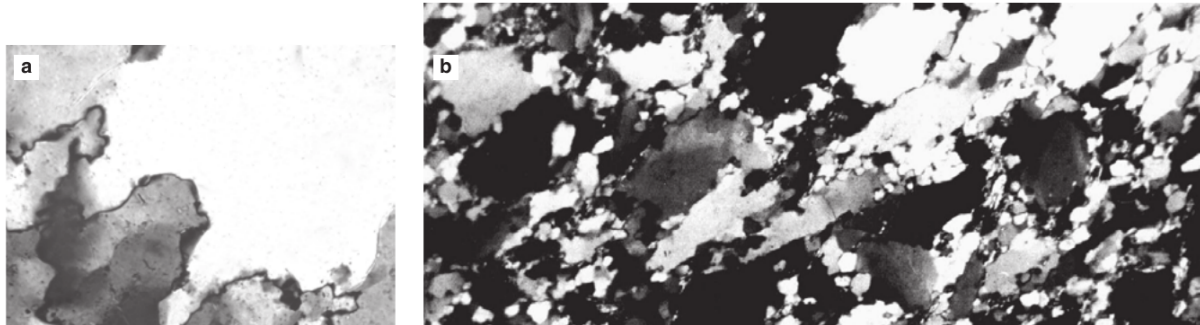


Figure 23-6 from John D. Winter (p. 510): Recrystallization by (a) grain-boundary migration and (b) subgrain rotation. Note how migration of grain boundaries eliminates jagged, irregular edges to form straight boundaries and larger, unstrained equant crystals.

2. The 120° Triple Junction & Foam Texture Analogy

In an isotropic, monomineralic aggregate (such as pure quartzite or calcite marble), all grain boundaries possess identical interfacial free energy ($\gamma_{11} = \gamma_{12} = \gamma_{13}$). At any line where three grains meet—termed a **triple junction**—interfacial surface tensions balance like forces in mechanical vector equilibrium: $\vec{\gamma}_1 + \vec{\gamma}_2 + \vec{\gamma}_3 = 0$. Because the magnitudes are equal, the three dihedral angles must be identically: $\theta = \frac{360^\circ}{3} = 120^\circ$

may commonly control reaction progress and explain why reaction rates are usually slightly slower than strain rates and faster in nature than in static lab experiments. Although strain may permit reactions to proceed toward equilibrium, it cannot change the nature of the equilibrium mineralogy, as was once proposed for “stress minerals” that were mistakenly believed to form only under conditions of deviatoric stress.

2 TEXTURES OF CONTACT METAMORPHISM

Contact metamorphism occurs in aureoles around intrusive bodies and is a response of cooler country rocks to the thermal $\pm$ metasomatic effects of the intrusion. The textures of contact metamorphism are typically developed at low pressures and thus under conditions of low deviatoric stress. The thermal maximum in a contact aureole should also occur much later than any stress imparted by forceful intrusion. Crystallization (including recrystallization) may therefore occur in a near-static environment, and contact metamorphism is typically characterized by a lack of significant preferred mineral orientation. Many minerals are equidimensional, rather than elongated, and the elongated minerals that do form are orientated randomly. Relict textures are also common in contact metamorphic rocks because there is little accompanying shear to destroy them.

Static recrystallization occurs after deformation ceases at elevated temperatures or when a thermal disturbance occurs in a low-stress environment. As described above, the process acts to reduce both lattice strain energy

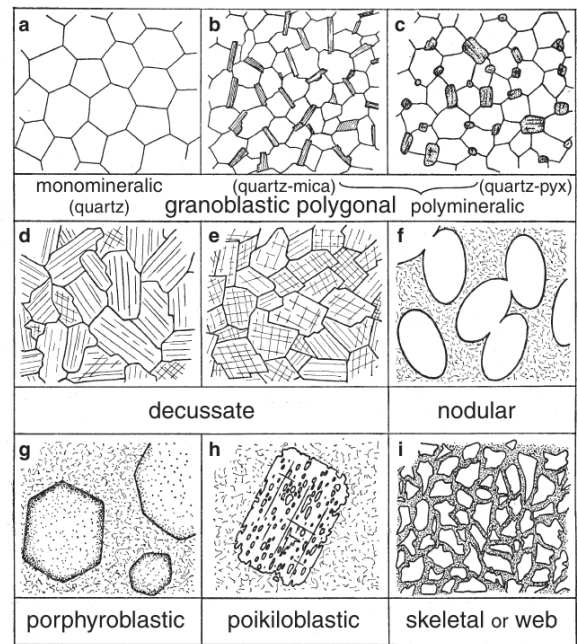


FIGURE 9 Typical textures of contact metamorphism. From Spry (1969).

surfaces become larger. To understand this, imagine a cube on a table. If the four vertical faces are of higher energy (less stable) and grow more quickly than the top and bottom faces, the cube will expand laterally faster than it does vertically, and it gradually becomes more plate-like. Low-energy surfaces in such minerals (e.g., the top and bottom of the plate)

Figure 23-9 from John D. Winter (p. 512): Typical textures of contact metamorphism and static annealing. (a) **Granoblastic polygonal texture** in marble/quartzite, showing equant grains with straight boundaries meeting at 120° triple junctions (the classic foam texture).

This arrangement is mathematically identical to a soap bubble froth (foam structure). A rock exhibiting equigranular, polygonal crystal outlines with planar boundaries intersecting at approximately 120° is universally designated as having a **granoblastic polygonal texture**.



Official Exam Question Statement & In-Depth Critical Analysis

Directorate General of Mines Safety (DGMS) 2019 · Junior Scientific Officer Exam · Question 10

Question Statement

Q.10: Granoblastic polygonal texture is an example of:

(a) Texture of recrystallization

(b) Texture of crystallization

(c) Texture of metasomatism

(d) Texture of mylonitization

Correct Answer: (a) Texture of recrystallization



Granoblastic polygonal texture is the quintessential product of solid-state static recrystallization (thermal annealing) driven by the minimization of grain boundary surface energy.

Exhaustive Option-by-Option Critical Analysis

(a) Texture of recrystallization

CORRECT

Why it is uniquely correct: In metamorphic geology and petrology (Winter, 2010, Chapter 23, pp. 510–512; Passchier & Trouw, 2005; Vernon, 2004), **granoblastic polygonal texture** (also called *polygonal mosaic texture* or *foam texture*) is universally recognized as the hallmark texture of **solid-state recrystallization**. During static thermal annealing following deformation (or during contact metamorphism in hornfelses and marbles), grain-boundary migration eliminates irregular, strained boundaries to minimize surface free energy, causing grain contacts to straighten into planar boundaries that meet at characteristic $\sim 120^\circ$ triple junctions.

(b) Texture of crystallization

INCORRECT

Why it is incorrect: A "texture of crystallization" specifically refers to **primary igneous crystallization** directly from a liquid silicate melt (producing textures such as ophitic, subophitic, porphyritic, trachytic, or cumulate textures with euhedral crystal faces) or primary sedimentary precipitation from an aqueous solution. Granoblastic polygonal texture forms strictly in the solid state by post-depositional/post-magmatic recrystallization.

(c) Texture of metasomatism

INCORRECT

Why it is incorrect: Textures of metasomatism are characterized by active, open-system chemical mass transfer driven by hydrothermal fluids. They produce **replacement textures**, reactionary overgrowths, compositional zoning, atoll structures, and vein fillings (such as in skarns and greisens), rather than equigranular, stable 120° polygonal mosaics.

(d) Texture of mylonitization

INCORRECT

Why it is incorrect: Mylonitization is a **high-strain dynamic deformation process** operating in ductile shear zones. It produces strong foliation, grain size reduction, ribbon quartz, rotated porphyroclasts (σ - and δ -type mantles), and core-and-mantle structures. Mylonitic textures represent high accumulated strain, whereas granoblastic polygonal texture represents the complete thermal destruction (annealing) of such strain!



Synthesis Reference Tables, Diagnostic Criteria & High-Yield Practice

Metamorphic Textural Genetic Categories & 8 Exam-Grade Multiple-Choice Questions

Classification of Metamorphic Textures by Genetic Mechanism

Genetic Texture Category	Primary Physical Mechanism	Defining Textural Features	Representative Rock Types
Texture of Recrystallization (Annealing)	Static solid-state grain boundary migration minimizing surface free energy	Granoblastic polygonal, 120° triple junctions, straight boundaries, equant grains	Marble, Quartzite, Granulite, Hornfels
Texture of Dynamic Deformation (Mylonitization)	Crystal-plastic dislocation creep and dynamic recrystallization during shear	Ribbon quartz, core-and-mantle, rotated porphyroclasts, ultramylonitic foliation	Mylonite, Ultramylonite, Phyllonite
Texture of Cataclasis (Brittle)	Brittle microfracturing, grain rolling, and frictional mechanical grinding	Angular clasts in fine comminuted matrix, lack of foliation (mortar texture)	Fault breccia, Cataclasite
Texture of Metasomatism	Fluid-mediated chemical replacement and mass transport	Pseudomorphs, compositional reaction rims, skarn bands, poikiloblastic veins	Skarn, Greisen, Fenite

8 High-Yield Practice Questions on Metamorphic Textures & Recrystallization

1. The dihedral angle of 120° at triple junctions in granoblastic polygonal textures results directly from:

- (a) Uniaxial compression
- (b) Balance of equal interfacial surface energies among three identical adjacent grains
- (c) Rapid quenching from magma
- (d) Dissolution along cleavage planes

Answer: (b) — In monomineralic aggregates, vector balance of equal interfacial surface tensions produces dihedral angles of 120° .

2. Which term describes the process where strained grains are replaced by new, strain-free grains during prolonged exposure to elevated temperatures without active deformation?

- (a) Cataclasis (b) Static annealing (recrystallization) (c) Dynamic fracturing (d) Exsolution

Answer: (b) Static annealing (recrystallization) — Post-kinematic heating allows grain boundaries to migrate and eliminate dislocations.

3. A monomineralic metamorphic rock composed entirely of interlocking equant quartz crystals displaying 120° triple junctions is classified texturally as:

- (a) Nematoblastic (b) Granoblastic polygonal quartzite (c) Lepidoblastic schist (d) Mylonitic ribbon

Answer: (b) Granoblastic polygonal quartzite — Granoblastic describes equant grains; polygonal denotes the straight-sided 120° geometry.

4. The term "foam texture" is frequently used as a synonym for which metamorphic texture?

- (a) Granoblastic polygonal (b) Slaty cleavage (c) Amygdaloidal (d) Porphyroclastic

Answer: (a) Granoblastic polygonal — Soap foam cells naturally meet at 120° angles to minimize surface tension, matching annealed crystals.

5. In contrast to static recrystallization, dynamic recrystallization occurs:

- (a) During active tectonic deformation and stress
(b) At absolute zero temperature
(c) Only in the presence of molten sulfur
(d) Exclusively on meteorites

Answer: (a) During active tectonic deformation and stress — Dynamic recrystallization proceeds synchronously with differential stress via subgrain rotation and bulging.

6. When unequal mineral phases (e.g., quartz and biotite) meet at a boundary, why do dihedral angles deviate from 120° ?

- (a) Interfacial energies between different mineral phases ($\gamma_{\text{Qtz-Bt}}$) differ from those between identical phases ($\gamma_{\text{Qtz-Qtz}}$)
(b) The rock ceases to be metamorphic
(c) Chemical equilibrium is impossible
(d) Gravity distorts the crystal lattice

Answer: (a) — Interfacial free energy depends on lattice misfit across differing mineral species, shifting dihedral angles away from 120°.

7. Which of the following rocks is most characteristically dominated by a granoblastic polygonal texture?

- (a) Pure Calcite Marble (b) Graphitic Slate (c) Chlorite Schist (d) Ultramylonite

Answer: (a) Pure Calcite Marble — Pure calcite recrystallizes readily under thermal metamorphism to form uniform granoblastic polygonal mosaics.

8. What happens to grain size during progressive static grain growth (Ostwald ripening) in an annealed metamorphic rock?

- (a) Mean grain size increases as larger grains consume smaller grains to reduce total grain boundary area
(b) Grain size reduces continuously to sub-micron size
(c) Grains dissolve into pore gas
(d) Grains convert to glass

Answer: (a) — Grain growth driven by surface energy reduction causes larger grains to grow at the expense of smaller grains with higher surface-to-volume ratios.